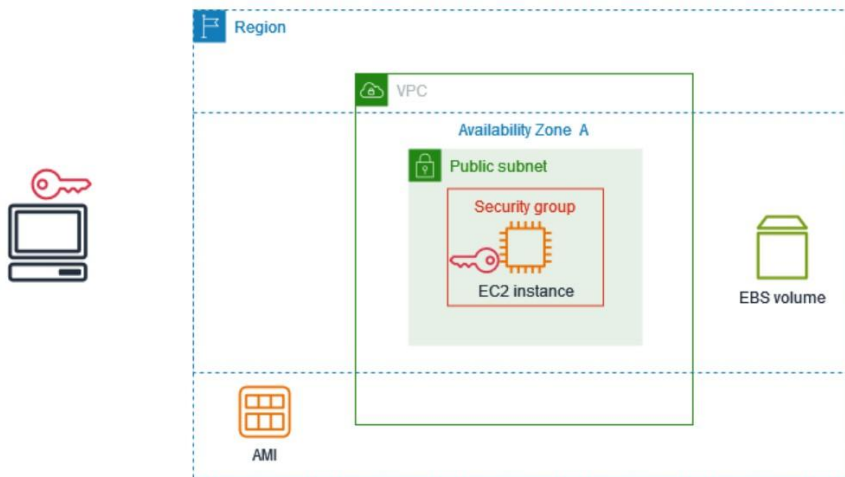


Lab : Launching a Linux Web Server using EC2 Instance

Lab overview

In this lab, you create an Amazon Elastic Compute Cloud (Amazon EC2) instance that hosts a simple website.



Access the AWS Management Console

1. To start the lab session, choose **Start Lab** in the upper-right corner of the page.

- o

- o The lab session starts.

A timer displays in the upper-right corner of the page and shows the time remaining in the session.

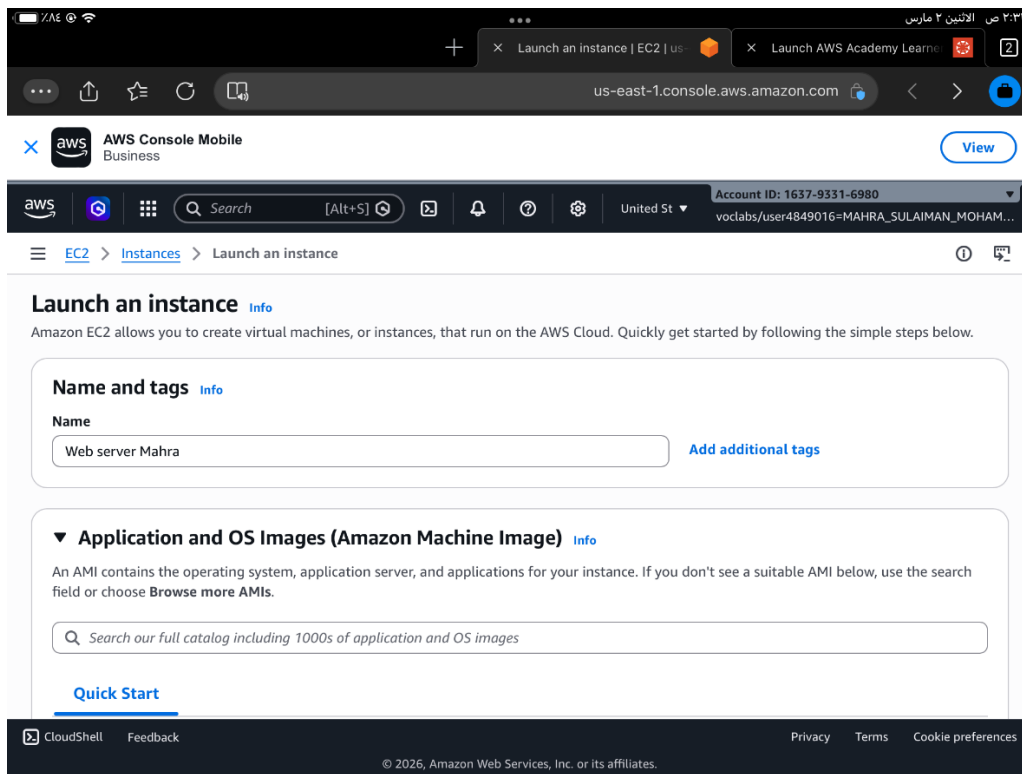
Tip: To refresh the session length at any time, choose **Start Lab** again before the timer reaches 0:00.

Before continuing, wait until the lab environment is ready. The environment is ready when the lab details appear on the right side of the page and the circle icon next to the **AWS** link in the upper-left corner turns green.

2. To return to these instructions, choose the **Readme** link in the upper-right corner.
3. To connect to the AWS Management Console, choose the **AWS** link in the upper-left corner, above the terminal window. A new browser tab opens and connects you to the AWS Management Console.

Task 1. Start creating the instance and assign a name

4. Choose the **Services** menu, locate the **Compute** services, and select **EC2**.
5. Choose the **Launch instance** button in the middle of the page, and then select **Launch instance** from the dropdown menu.
6. Name the instance:
 - o Give it the name Web Server 1

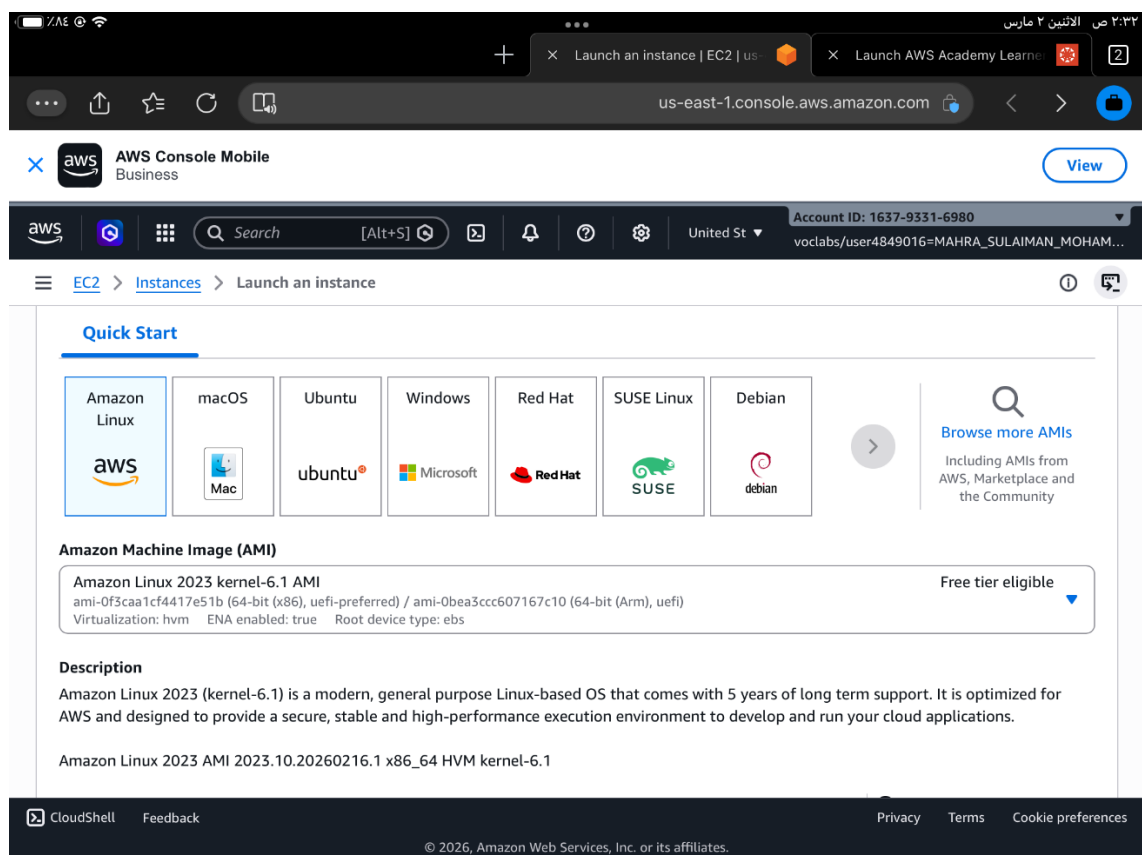


Task 2. Application and OS Images

7. Choose an AMI from which to create the instance:

-
- In the list of available *Quick Start* AMIs, keep the default **Amazon Linux** AMI selected. Then choose **Amazon Linux 2 AMI (HVM)** free tier from the list.

The type of *Amazon Machine Image (AMI)* you choose determines the Operating System (OS) that will run on the EC2 instance that you launch. In this case, you have chosen Amazon Linux 2 as the guest OS.

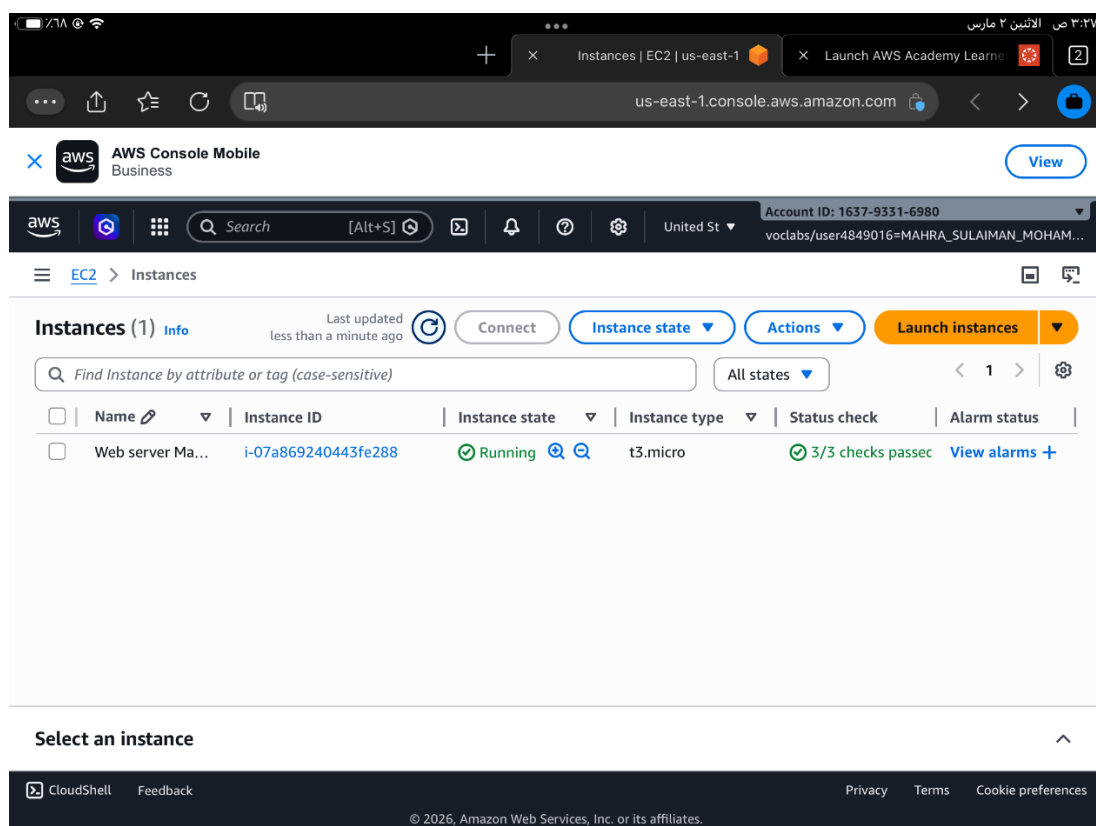


Task 3. Choose an instance type

8. Specify an Instance type:

- In the *Instance type* panel, keep the default **t2.micro** selected.

The *Instance Type* defines the hardware resources assigned to the instance. This instance type has 1 virtual central processing unit (CPU) and 1 GiB of memory.

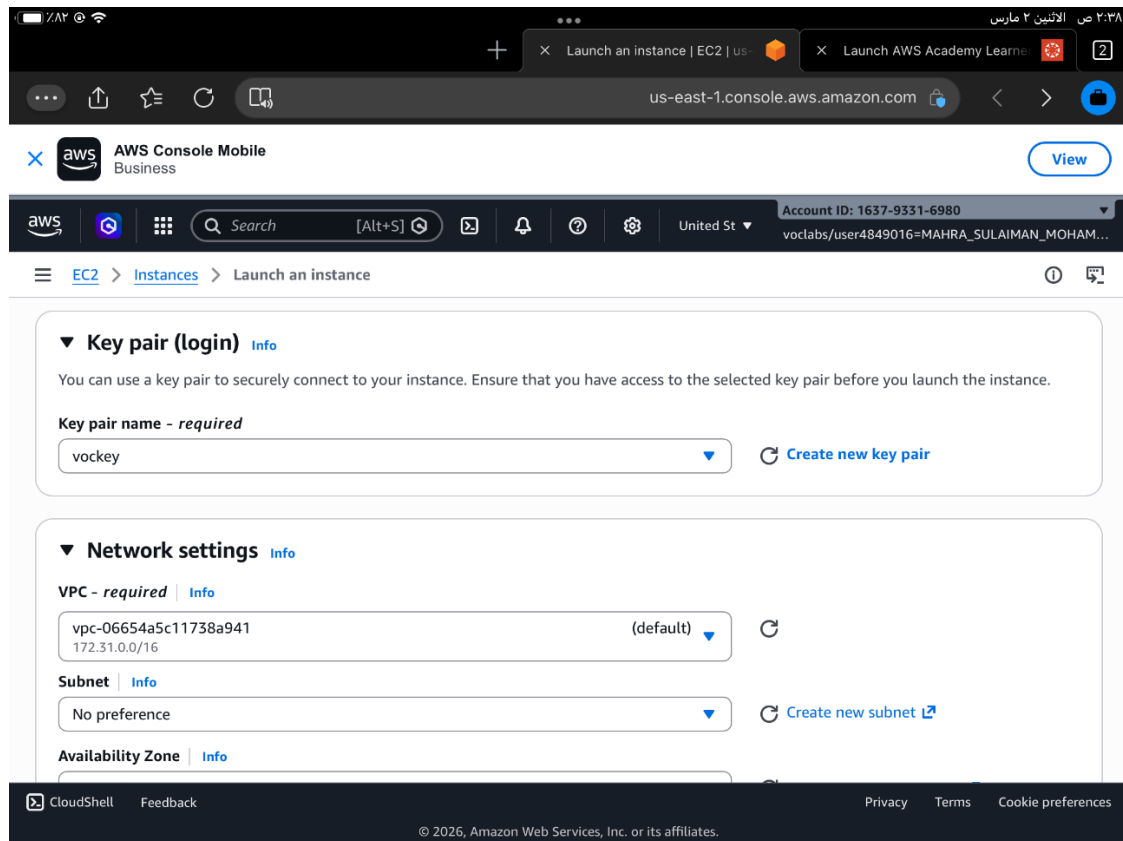


Task 4. Choose a key pair

9. Select the key pair to associate with the instance:

- From the **Key pair name** menu, select **vockey** or you can create a new key pair.

The *vockey* key pair you selected will allow you to connect to this instance via SSH after it has launched. Although you will not need to do that in this lab, it is still required to identify an existing key pair, or create a new one, when you launch an instance.



Task 5. Network settings

10. Next to Network settings, choose **Edit**.
11. Keep the default *VPC* and *subnet* settings. Also keep the **Auto-assign public IP** setting set to **Enable**.

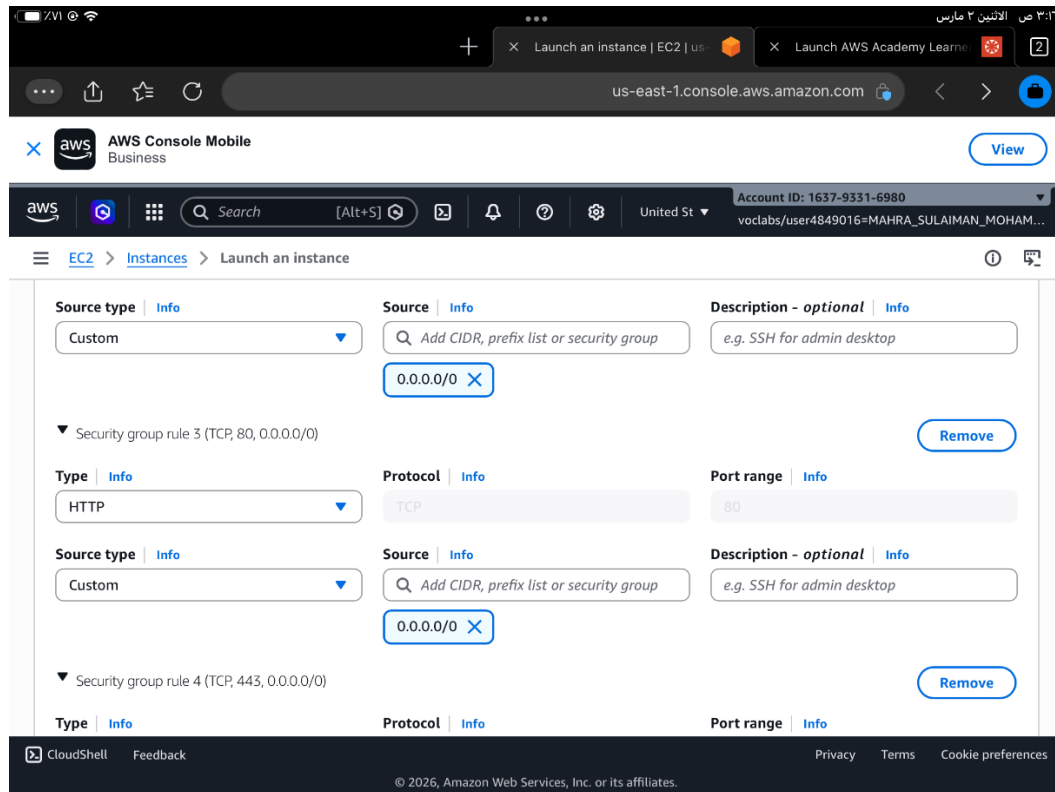
The Network indicates the virtual private cloud (VPC) you want to launch the instance into..

12. Under *Firewall (security groups)*, keep the default **Create security group** option chosen.
13. Configure a new security group:

- Keep the default selection **Create a new security group**.
- **Security group name:** Clear the text and enter Web Server
- **Description:** Clear the text and enter Security group for my web server
- Keep the default selection SSH inbound rule.
- Allow HTTPS traffic from the internet by selecting the check box.
- Allow HTTP traffic from the internet by selecting the check box.

A *security group* acts as a virtual firewall that controls the traffic for one or more instances.

When you launch an instance, you associate one or more security groups with the instance. You add rules to each security group that allow traffic to or from its associated instances. You can modify the rules for a security group at any time. The new rules are automatically applied to all instances that are associated with the security group.



Task 6. Configure storage

14. In the *Configure storage* section, keep the default settings.

You will launch the Amazon EC2 instance using a default Elastic Block Store (EBS) disk volume. This will be your root volume (also known as a *boot volume*) which will host the Amazon Linux 2022 guest operating system that you specified earlier. It will run on a general purpose SSD (*gp2*) hard drive that is 8 GiB in size. You could alternatively add more storage volumes, however that is not needed in this lab.

Task 7. Advanced details

15. Configure a script to run on the instance when it launches:

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- o Expand the **Advanced details** panel.

Scroll to the bottom of the page and then copy and paste the code shown below into the **User data** box:

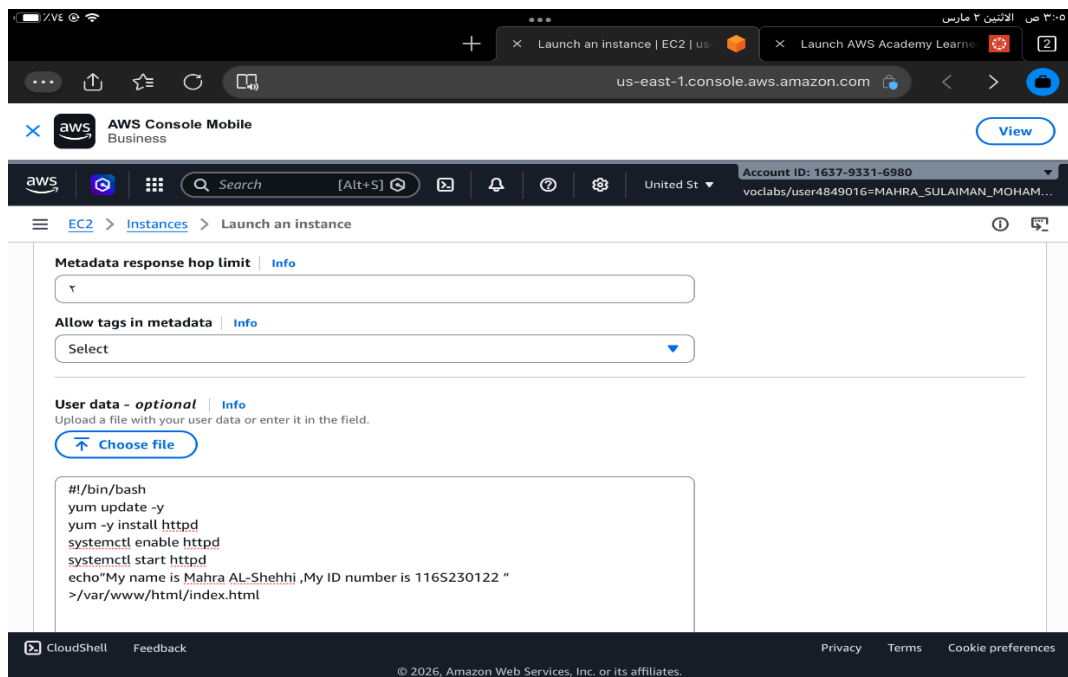
```
#!/bin/bash yum update -y yum -y install httpd
```

```
systemctl enable httpd systemctl start httpd echo
```

```
“Hello Cloud” > /var/www/html/index.html
```

This bash script will run with root user permissions on the guest OS of the instance. It will run automatically when the instance launches for the first time. This script does the following:

- Updates the server
 - Installs an Apache web server (httpd)
 - Configures the web server to automatically start on boot
 - Activates the web server
- Creates a simple webpage



Task 8. Review the instance and launch

16. At the bottom of the **Summary** panel on the right side of the screen choose Launch instance You will see a Success message.
17. Choose View all instances

CSCL2101: Cloud Computing Essentials

The instance will first appear in the *Pending* state, which means it is being launched. The state will then change to *Running*, which indicates that the instance has started booting. It takes a few minutes for the instance to boot.

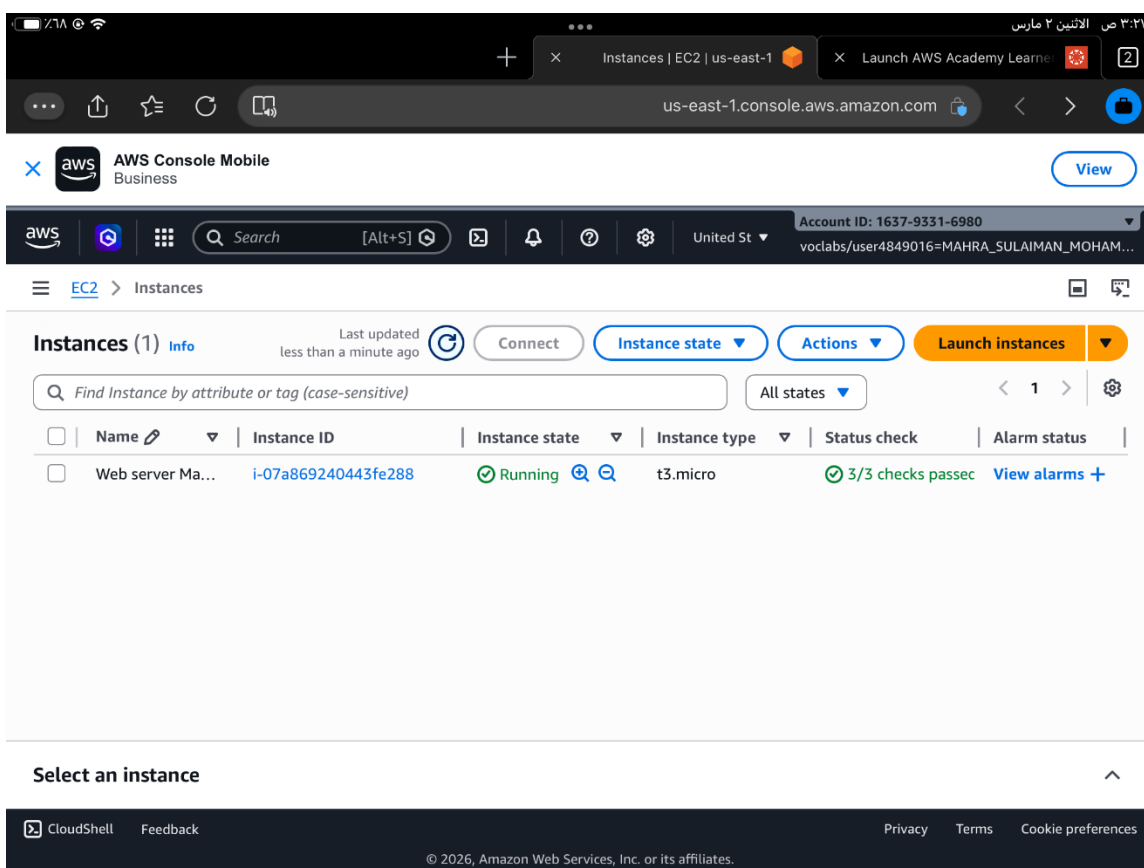
18. Select the **Web Server 1** instance, and review the information in the **Details** tab that displays in the lower pane.

Notice that the instance has a **Public IPv4 address**. You can use this IP address to communicate with the instance from the internet.

19. Before you continue, wait for your instance to display the following:

-
- **Instance state:** *Running*
Status check: *2/2 checks passed*

This may take a few minutes. Choose the refresh icon at the top of the page every 30 seconds or so to more quickly become aware of the latest status of the instance.



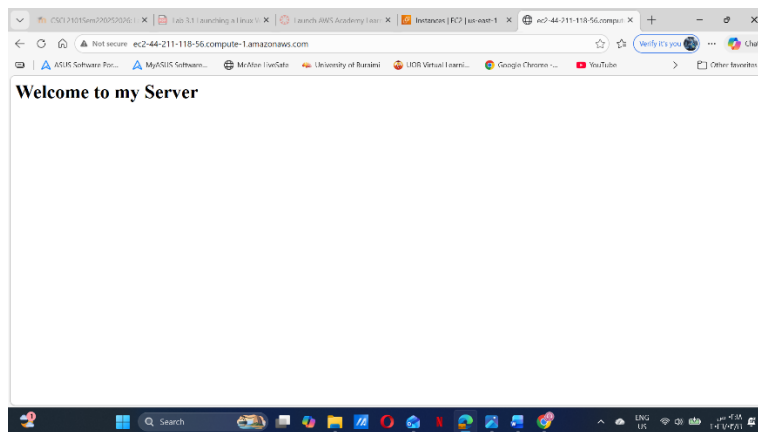
Task 9. Access your EC2 instance

When you launched your EC2 instance, you provided a script that installed a web server and created a simple webpage. In this task, you will try to access the content from the web server.

20. From the **Details** tab, copy the **Public IPv4 DNS** value of your instance to your clipboard.

Note: A *public* address means that the instance can be reached from the internet. Each instance that receives a public IP address is also given an external DNS hostname; for example, ec2-xxxxxx-xxx-xxx.compute1.amazonaws.com. AWS resolves an external DNS hostname to the *public* IP address of the instance when communication comes from outside its VPC. When communication comes from inside its VPC, the DNS hostname is resolved to the *private* IPv4 address.

21. Open a new tab in your web browser, paste the **Public IPv4 DNS** you just copied, and press **Enter**.



I opened a new browser tab and entered the public IPv4 DNS address. The webpage loaded successfully and displayed the “Welcome to my Server” message confirming that the Apache web server is running and accessible.

Lab complete

Congratulations! You have completed the lab. 22.

Log out of the AWS Management Console.

- In the
 - upper-right corner of the page, choose your user name. Your user name begins with **voclabs/user**.
 - Choose **Sign out**. 23. Choose **End Lab** at the top of this page, and then select **Yes** to confirm that you want to end the lab.

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